

USAL

جامعة العلوم والآداب اللبنانية

UNIVERSITY OF SCIENCES & ARTS IN LEBANON

Topic 1:
Equations and Inequalities

A- Linear equation of first degree

1. Definition

Let a and b be two real numbers with **a ≠ 0** and x an unknown to be determined.

A first-degree equation is any expression that can be simplified to the form of **ax + b = 0**

a and **b** are called the **coefficients** of the equation .

To solve the equation **ax + b = 0** is to find the value of x which verifies it. The **solution** or the **root** of this equation is

$$\mathbf{x = -\frac{b}{a}}$$

Example 1

- 4 is a solution of the equation 2x – 8 = 0, but 3 is not a solution of it.
- Solve the following equations: $4(2x-1) = 5x + 8$

$$3(2x-1) = 2(3x+4)$$

$$6(-3x+2) = -3(6x-4)$$

2. Reduction of an equation into a linear equation

a) Equation that can be factorized

There are equations that are not of the first degree but that can be written as a product of first-degree factors equal to zero.

If ***A*** and ***B*** are real numbers, $A.B = 0$ is equivalent to $A = 0$ or $B = 0$

Example 2

Solve the following equations

- $(2x - 3)(3x + 4) = 0$
- $\mathrm{x}(3x - 5)(2x + 7) = 0$
- $x^2 - 1 = (x + 1)(2x + 5)$

b) Equations having the unknown in the denominator

There are equations that have the unknown in their denominators.

If *A* and *B* are real numbers, $\frac{A}{B} = 0$ is equivalent to $A = 0$ and $B \neq 0$

Example 3

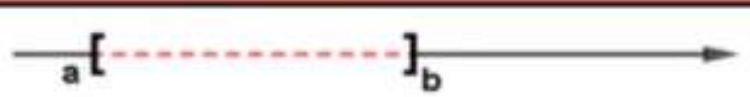
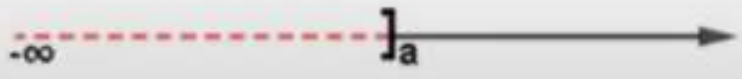
Solve the following equations

- $\frac{x-1}{x} = 0$
- $\frac{x-3}{x-1} = 0$
- $\frac{3}{x^2-9} + \frac{2}{x+3} = \frac{1}{x-3}$

B- Linear inequality of first degree

Recall

Represent an interval on a number line

Interval	Representation
$[a ; b]$	
$]a ; b[$	
$]a ; b]$	
$[a ; b[$	
$[a ; +\infty[$	
$]a ; +\infty[$	
$]-\infty ; a]$	
$]-\infty ; a[$	
$\mathbb{R} =]-\infty ; +\infty[$	

B- Linear inequality of first degree

1. Definition

Let a and b be two real numbers, a ≠ 0, and x an unknown.

A first-degree inequality in one unknown is any writing of the form:

$ax + b \geq 0$, $ax + b > 0$, $ax + b \leq 0$ or $ax + b < 0$.

a and **b** are called the coefficients of the inequality.

To solve the inequality is to find the value of x which verifies it.

Example 1

Solve the following inequalities

- $3x - 9 \leq 0$
- $\frac{1}{2}x - 3 < x + 2$
- $2x - 3 < \frac{5x-1}{4} - 5$

2. Inequality of the form $m < ax + b < M$

It is equivalent to the system of two inequalities $\begin{cases} m < ax + b \\ ax + b < M \end{cases}$.

This system can be solved directly by applying the following properties:

Let a, b, and c be 3 non zero real numbers, such that $a < x < b$, so

- $a + c < x + c < b + c$
- If $c > 0$ then $ac < xc < bc$
- If $c < 0$ then $bc < xc < ac$

Example 2

Solve in $\mathbb{R}$ the inequality $2 < 3x - 2 < 5$

3. Framing a polynomial

Conversely if we use **Property 1** mentioned in the slide before we can frame

a polynomial of the form $ax + b$, where a and b are two real numbers and a $\neq$ 0.

Example 3

Given that $1 < x < 2$, frame $-2x + 4$

C- Absolute value

1. Definition

On an x′Ox axis, x is the abscissa of a point M.

The absolute value of x, denoted by |x|, is the distance OM.

So OM = |x|

$$|x| = \begin{cases} +x & \text{if } x \geq 0 \\ -x & \text{if } x \leq 0 \end{cases}$$

Example 1

- $|\sqrt{3} - 1| = \sqrt{3} - 1$
- $|\sqrt{2} - 2| = -(\sqrt{2} - 2)$

2. Simplifying expressions containing absolute value

Properties:

- $|-x| = |x|$
- $|x - y| = |y - x|$
- $|x.y| = |x|.|y|$
- $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}, y \neq 0$
- $|x| = |y|$ is equivalent to $x = y$ or $x = -y$
- $|x + y| \leq |x| + |y|$
- $|x - y| \leq |x| + |y|$

3. Solving equations with absolute value

Let a be a real number

- If $a > 0$, and $|x| = a$ then $x = a$ or $x = -a$
- If $a < 0$, and $|x| = a$ then no solution
- If $a = 0$, and $|x| = a$ then $x = 0$

Example 2

Solve the following equations

- $|3x + 2| = 4$
- $|x + 2| + 3 = 1$
- $|2x + 4| = 0$

4. Solving inequalities with absolute value

Let a be a real number

a) Inequality of the form $|x| \leq a$

- If $a > 0$, and $|x| \leq a$ then $-a \leq x \leq a$
- If $a < 0$, and $|x| \leq a$ then no solution
- If $a = 0$, and $|x| \leq a$ then $x = 0$

b) Inequality of the form $|x| \geq a$

- If $a > 0$, and $|x| \geq a$ then $x \geq a$ or $x \leq -a$
- If $a \leq 0$, and $|x| \geq a$ then any value of $\mathbb{R}$ is a solution of this inequality

Example 3

Solve the following inequalities

- $|x - 1| < 2$
- $|x + 2| \geq 3$
- $|2x + 4| \leq 0$
- $|x + 1| + 3 \geq 1$
- $|2x + 3| - 2 \leq -6$

D- Quadratic equation

1. Definition

- The expression $ax^2 + bx + c$ is called polynomial of the second degree or **trinomial** of the second degree if a $\neq$ 0.

  a is the coefficient of x^2 , b that of x and c is the constant term.
- The equation $\boldsymbol{ax^2 + bx + c = 0}$, with a non-zero, is called **equation of the second degree in x**. a , b and c are numerical or parametric coefficients.
- Any value of the unknown x that verifies the equation $\boldsymbol{ax^2 + bx + c = 0}$, is called **solution** or **root** of this equation.

2. Solution of second degree equation

Let a, b, and c be three real numbers such that $a \neq 0$, and (E): $ax^2 + bx + c = 0$.

Consider $\Delta = b^2 - 4ac$

- If $\Delta < 0$, then the equation (E) has no real solutions
- If $\Delta = 0$, then the equation (E) has a double root $x_1 = x_2 = -\frac{b}{2a}$
- If $\Delta > 0$, then the equation (E) has two real solutions: $x_1 = \frac{-b-\sqrt{\Delta}}{2a}$, and $x_2 = \frac{-b+\sqrt{\Delta}}{2a}$

Example 1

Solve the following equations

- $3x^2 - 4x + 1 = 0$
- $4x^2 - 12x + 9 = 0$
- $2x^2 - 7x + 8 = 0$

3. Change of variable

An example of the change of variable is the bisquared equation.

It is of the form $ax^4 + bx^2 + c = 0$, where $a \neq 0$.

To solve it, we introduce an auxiliary unknown, supposing, for example $t = x^2$.

The equation in t is : **$at^2 + bt + c = 0$** .

We calculate the value of t if it exists then we deduce x.

Exapmle 2

Solve the bisquared equation

$$3x^4 + 5x^2 - 8 = 0$$

E- Sign of a polynomial

I- Sign of a first-degree polynomial

Let F(x) = ax + b where a, and b be two real numbers such that $a \neq 0$.

$a > 0$	x: $-\infty$; $-\frac{b}{a}$; $+\infty$ $F(x)$: $-$; 0 ; $+$
$a < 0$	x: $-\infty$; $-\frac{b}{a}$; $+\infty$ $F(x)$: $+$; 0 ; $-$

II- Sign of a second-degree polynomial

Let $F(x) = ax^2 + bx + c$ where a, b, and c be three real numbers such that $a \neq 0$.

<table>
<tr><td>$\Delta<0$</td><td>No solutions</td><td>$a > 0$</td><td>$F(x) > 0$</td></tr>
<tr><td></td><td></td><td>$a < 0$</td><td>$F(x) < 0$</td></tr>
<tr><td>$\Delta=0$</td><td>One double root</td><td>$a > 0$</td><td>$F(x) \geq 0$</td></tr>
<tr><td></td><td></td><td>$a < 0$</td><td>$F(x) \leq 0$</td></tr>
<tr><td>$\Delta>0$</td><td>Two distinct roots</td><td>$a > 0$</td><td>
<table>
<tr><td>x</td><td>$-\infty$</td><td>X_1</td><td></td><td>X_2</td><td>$+\infty$</td></tr>
<tr><td>F(x)</td><td>+</td><td>0</td><td>−</td><td>0</td><td>+</td></tr>
</table>
</td></tr>
<tr><td></td><td></td><td>$a < 0$</td><td>
<table>
<tr><td>x</td><td>$-\infty$</td><td>X_1</td><td></td><td>X_2</td><td>$+\infty$</td></tr>
<tr><td>F(x)</td><td>−</td><td>0</td><td>+</td><td>0</td><td>−</td></tr>
</table>
</td></tr>
</table>

Application:

- Study the sign of f(x) in each of the following cases:

$f(x) = 2x + 1$

x	$-\infty$		$-\frac{1}{2}$		$+\infty$
f(x)		$-$	0	$+$	

$f(x) = -4x + 8$

x	$-\infty$		2		$+\infty$
f(x)		$+$	0	$-$	

$f(x) = 4x^2 - 2x - 2$

$\Delta > 0$

x	$-\infty$		-1/2		1		$+\infty$
f(x)		$+$	0	$-$	0	$+$	

$f(x) = 4x^2 - 12x + 22$

$\Delta < 0$

x	$-\infty$		$+\infty$
f(x)		$+$	

$f(x) = -x^2 + 12x - 36$

$\Delta = 0$

x	$-\infty$		6		$+\infty$
f(x)		$-$	0	$-$	

$f(x) = \frac{2x^2-3x+1}{x-3}$

x	$-\infty$		½		1		3		$+\infty$
$2x^2-3x+1$		+	0	−	0	+	|	+	
$x-3$		−	|	−	|	−	0	+	
f (x)		−	0	+	0	−	||	+	

So $f(x) \geq 0$ for $x \in [1/2\ ,\ 1] \cup\ ]\ 3,\ +\infty[$

$f(x) = \frac{x^2-7x+22}{2x^2-8}$

x	$-\infty$		-2		2		$+\infty$
$x^2-7x+22$		+	|	+	|	+	
$2x^2-8$		+	0	−	0	+	
f (x)		+	||	−	||	+	

So $f(x) < 0$ for $x \in\ ]\ -2,\ 2[$

$f(x) = \frac{-x^2+5x-6}{(x-1)^2}$

x	$-\infty$		1		2		3		$+\infty$
$-x^2+5x-6$		−	|	−	0	+	0	−	
$(x-1)^2$		+	0	+	|	+	|	+	
f (x)		−	||	−	0	+	0	−	

So $f(x) \geq 0$ for $x \in [2,\ 3]$